New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code: BE01R00041

Module 5: Multivariable Calculus

Weightage: 20%

Topics: Center of mass & Gravity:
(Constant & Variable densities) :-

→ Using a single integral we able to compute the center of mass for a one dimensional object with variable density & two dimensional object with constant density

→ With help of Double integration, we can handle two dimensional & variable density.

→ The co-ordinate of center of mass. are,

$$\bar{X} = \frac{M_y}{M} \qquad \& \qquad \bar{Y} = \frac{M_x}{M}$$

where M → Total mass
M_x → The moment around the x-axis.
M_y → The moment around the y-axis

→ **Center of Gravity is** $(\bar{x}, \bar{y})$.
In 2-dimentional case, we compute M, M_y & M_x using the formula

$$M = \int_{x_0}^{x_1} \int_{y_0}^{y_1} \sigma(x,y)dydx, \text{ where } \sigma(x,y) \text{ is density}$$

$$M_x = \int_{x_0}^{x_1} \int_{y_0}^{y_1} \boldsymbol{y} \cdot \sigma(x,y)dy\, dx$$

$$M_y = \int_{x_0}^{x_1} \int_{y_0}^{y_1} \boldsymbol{x} \cdot \sigma(x,y)dy\, dx.$$

Ex-1 : Find the center of mass of a two dimensional plate that occupies the triangle

$0 \le x \le 1, 0 \le y \le x$, & has density function xy.

Solution: $\sigma(x, y) = xy$ (Density function given)

First find M, M_y & M_x

$$M = \int_{x=0}^{1} \int_{y=0}^{x} \sigma(x, y) dy\, dx$$

$$= \int_{0}^{1} \int_{0}^{x} xy\, dy\, dx$$

$$= \int_{0}^{1} x \left(\frac{y^2}{2}\right)_0^x dx$$

$$= \int_{0}^{1} \frac{x^3}{2} dx = \frac{1}{2}\left(\frac{x^4}{4}\right)_0^1 = \frac{1}{8}$$

$$M_x = \int_{x=0}^{1} \int_{y=0}^{x} y \cdot \sigma(x, y) dy\, dx$$

$$= \int_{0}^{1} \int_{0}^{x} y \cdot (xy) dy\, dx$$

$$= \int_{0}^{1} x \left(\frac{y^3}{3}\right)_0^x dx$$

$$= \int_{0}^{1} \frac{x^4}{3} dx = \frac{1}{3}\left(\frac{x^5}{5}\right)_0^1 = \frac{1}{15}$$

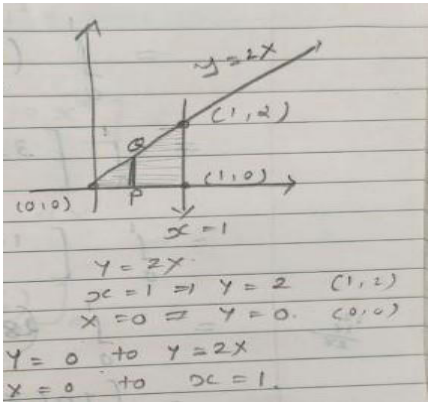
$$\rightarrow\ M_y = \int_{x=0}^{1}\int_{y=0}^{x} x\cdot\sigma(x,y)dydx$$
$$= \int_{0}^{1}\int_{0}^{x} x(xy)\ dydx$$
$$= \int_{0}^{1} x^2\left[\frac{y^2}{2}\right]_0^x dx$$
$$= \int_{0}^{1} \frac{x^4}{2}dx$$
$$= \frac{1}{2}\left(\frac{x^5}{5}\right)_0^1 = \frac{1}{10}$$

∴ Center of mass

$$\bar{X} = \frac{M_y}{M} = \frac{1/10}{1/8} = \frac{8}{10} = \frac{4}{5}$$
$$\bar{Y} = \frac{M_x}{M} = \frac{1/15}{1/8} = \frac{8}{15}$$
$$\therefore \text{Center of Gravity } (\bar{x},\bar{y}) = \left(\frac{4}{5},\frac{8}{15}\right).$$

Ex-2 : A thin plate covers the triangular region bounded by x-axis & the line $x = 1$ & $y = 2x$ in first quadrant. The plate density at the pt. (x, y) in $\sigma(x, y) = 6x + 6y + 6$ Find the plate's mass, first moments & centre of mass about co-ordinate axis.

Solution: Here $\sigma(x, y) = 6x + 6y + 6$



$$M = \int_{x=0}^{1}\int_{y=0}^{2x} \sigma(x,y)\, dydx$$

$$= \int_{0}^{1}\int_{0}^{2x} (6x + 6y + 6) dydx$$

$$= \int_{0}^{1} \left[6xy + 6\frac{y^2}{2} + 6y\right]_0^{2x} dx$$

$$= \int_{0}^{1} [6x(2x) + 3(2x)^2 + 6(2x)] dx$$

$$= \int_{0}^{1} [12x^2 + 12x^2 + 12x] dx$$

$$= 12\left[\frac{x^3}{3} + \frac{x^3}{3} + \frac{x^2}{2}\right]_0^1$$

$$= 12\left(\frac{2}{3} + \frac{1}{2}\right) = \left(\frac{4+3}{6}\right)$$

$$= 2(7) = 14$$

Moment about x-axis.

$$\Rightarrow M_x = \int_{x=0}^{1}\int_{y=0}^{2x} y \cdot \sigma(x,y)\, dydx$$

$$= \int_{0}^{1}\int_{0}^{2x} y \cdot (6x + 6y + 6)\, dydx$$

$$= \int_{0}^{1}\int_{0}^{2x} (6xy + 6y^2 + 6y)\, dydx$$

$$= \int_{0}^{1} \left[6x \cdot \frac{y^2}{2} + 6 \cdot \frac{y^3}{3} + 6y^2\right]_0^{2x} dx$$

$$= \int_0^1 [3xy^2 + 2y^3 + 3y^2]_0^{2x} dx$$

$$= \int_0^1 [3x(2x)^2 + 2(2x)^3 + 3(2x)^2] dx$$

$$= \int_0^1 [12x^3 + 16x^3 + 12x^2) dx$$

$$= \int_0^1 (28x^3 + 12x^2) dx$$

$$= \left[28\left(\frac{x^4}{4}\right) + 12\left(\frac{x^3}{3}\right)\right]_0^1$$

$$= (7 + 4) = 11$$

Moment about y-axis.

$$M_y = \int_0^1 \int_0^{2x} x \cdot \sigma(x, y) dy dx$$

$$= \int_0^1 \int_0^{2x} x(6x + 6y + 6) dy dx$$

$$= \int_0^1 \int_0^{2x} (6x^2 + 6xy + 6x) dy dx$$

$$= \int_0^1 \left(6x^2 y + 6x \cdot \frac{y^2}{2} + 6xy\right)_0^{2x} dx$$

$$= \int_0^1 [6x^2(2x) + 3x(2x)^2 + 6x(2x)] dx$$

$$= \int_0^1 [12x^3 + 12x^3 + 12x^2] dx$$

$$= \int_0^1 (24x^3 + 12x^2)dx$$

$$= \left[24\left(\frac{x^4}{4}\right) + 12\left(\frac{x^3}{3}\right)\right]_0^1$$

$$= (6 + 4)$$

$$= 10$$

∴ Center of mass

$$\bar{x} = \frac{M_y}{M} = \frac{10}{14} = \frac{5}{7}$$

$$\bar{y} = \frac{M_x}{M} = \frac{11}{14}$$

∴ Center of Gravity: $(\bar{x}, \bar{y}) = \left(\frac{5}{7}, \frac{11}{14}\right)$.

Ex-3 : Find the center of mass of a thin uniform plate whose shape is the region between $y = \cos x$ & x-axis. between $x = -\pi/2$ & $x = \pi/2$. Take the density if constant $\sigma(x, y) = 1$

$$m = \int_{-\pi/2}^{\pi/2} \int_0^{\cos x} 1dydx = 2$$

$$mx = \pi/4$$

$$my = 0$$

$$\therefore \bar{x} = 0/2 = 0$$

$$\bar{y} = 17/4/2 = 778$$

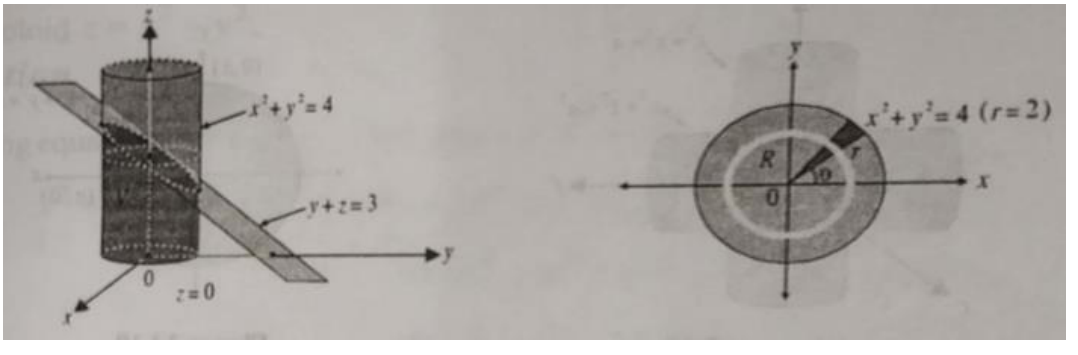
∴ center of Gravity $(\bar{x}, \bar{y}) = \left(0, \frac{\pi}{8}\right)$

Topics: **Applications of definite integral.**

❖ Applications of Triple Integrals (Volumes)

Example 1: Find the volume bounded by cylinder $x^2 + y^2 = 4$ and the plane $y + z = 3$ and $z = 0$.

Solution: the given solid region V in left figure by dark shaded portion and its projection on the xy- plane is a circle shown in right figure.



The required volume V is given by $V = \iint_R z \cdot r\, dr d\theta$

Here $z = 3 - y = 3 - r\sin\theta$

$$\therefore V = \int_{\theta=0}^{2\pi}\int_{r=0}^{2} (3 - r\sin\theta)\, r\, dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(3\frac{r^2}{2} - \sin\theta\frac{r^3}{3}\right)_0^2 d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(6 - \frac{8}{3}\sin\theta\right) d\theta$$

$$= \left(6\theta + \frac{8}{3}\cos\theta\right)_0^{2\pi}$$

$$= \left(12\pi + \frac{8}{3}\right) - \left(0 + \frac{8}{3}\right)$$

$$= 12\pi$$

Example 2: Find the volume below the surface $z = x^2 + y^2$, above the plane $z = 0$ and inside the cylinder $x^2 + y^2 = 2y$.

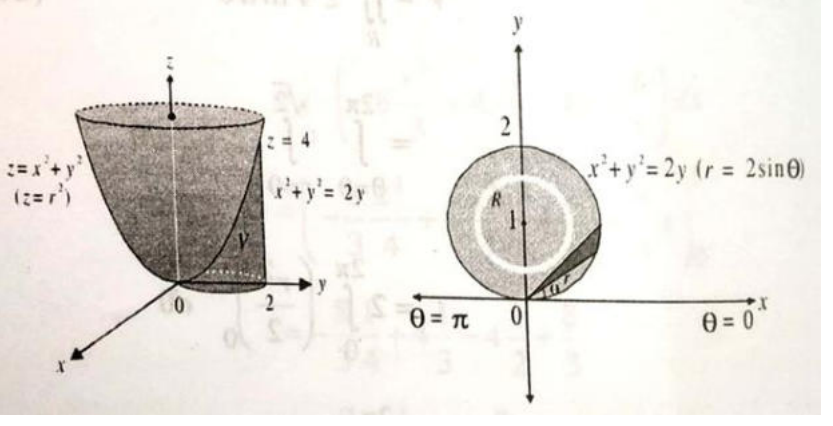
Solution: compare the eq^n $x^2 + y^2 = 2y$ with $x^2 + y^2 + 2gx + 2fy + c = 0$

$$2g = 0 \rightarrow g = 0, \quad 2f = -2 \rightarrow f = -1, \quad c = 0$$

$$\therefore center = (-g, -f) = (0,1)$$

$$radius = \sqrt{g^2 + f^2 - c} = 1$$

Solving equations $z = x^2 + y^2$ and $x^2 + y^2 = 2y$. We get $z = 2y$.



The required volume V is given by $V = \iint_R z \cdot r \, dr d\theta$

$$= \int_{\theta=0}^{\pi} \int_{r=0}^{2\sin\theta} (x^2 + y^2)\, r \, dr d\theta$$

$$= \int_{\theta=0}^{\pi} \int_{r=0}^{2\sin\theta} r^2 \, r \, dr d\theta$$

$$= \int_{\theta=0}^{\pi} \int_{r=0}^{2\sin\theta} r^3 \, dr d\theta$$

$$= \int_0^{\pi} \left(\frac{r^4}{4}\right)_0^{2\sin\theta} d\theta$$

$$= \frac{1}{4}\int_0^{\pi} 16\, sin^4\theta\, d\theta$$

$$= 4\int_0^{\pi} \left(\frac{1-\cos 2\theta}{2}\right)^2 d\theta \qquad since\ cos2\theta = 1 - 2sin^2\theta$$

$$= \int_0^{\pi} (1 - 2\cos 2\theta + cos^2 2\theta)\, d\theta$$

$$= \int_0^{\pi} (1 - 2\cos 2\theta + 1 - sin^2 2\theta)\, d\theta$$

$$= \int_0^{\pi} \left(2 - 2\cos 2\theta - \left(\frac{1-\cos 4\theta}{2}\right)\right) d\theta$$

$$= \frac{1}{2}\int_0^{\pi} (4 - 4\cos 2\theta - 1 + \cos 4\theta)\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi} (3 - 4\cos 2\theta + \cos 4\theta)\, d\theta$$

$$= \frac{1}{2}\left[3\theta - 4\left(\frac{\sin 2\theta}{2}\right) + \frac{\sin 4\theta}{4}\right]_0^{\pi}$$

$$= \frac{1}{2}\left(3\pi - 2\sin 2\pi + \frac{\sin 4\pi}{4}\right)$$

$$= \frac{1}{2}\left(3\pi - 2(0) + \frac{0}{4}\right)$$

$$= \frac{3\pi}{2}$$

Example 3: Find the volume of the region that lies under the paraboloid $z = x^2 + y^2$ and above the triangle enclosed by the lines $y = x, x = 0$ and $x + y = 2$ in the xy- plane.

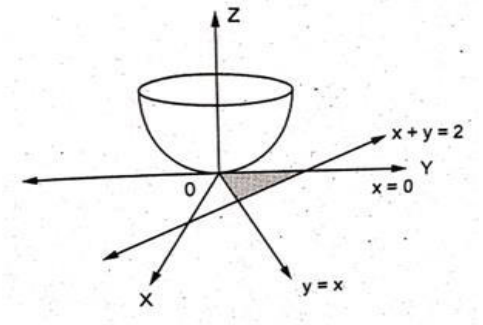
Solution: let $V = \iiint_R \; dv$

Here upper surface is $z = x^2 + y^2$ and lower surface is xy- plane. (i.e. $z = 0$)

$$\therefore V = \iint_S \int_{z=0}^{x^2+y^2} dz \; dA$$

$$= \iint_S (x^2 + y^2)\, dx\, dy$$

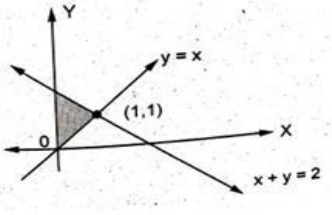
$$= \int_{x=0}^{1} \int_{y=x}^{y=2-x} (x^2 + y^2)\, dx\, dy$$



$$= \int_0^1 \left[x^2 y + \frac{y^3}{3} \right]_x^{2-x} dx$$

$$= \int_0^1 \left[x^2(2-x) + \frac{(2-x)^3}{3} - x^3 - \frac{x^3}{3} \right] dx$$

$$= \int_0^1 \left[(2x^2 - x^3) + \frac{(2-x)^3}{3} - \frac{4x^3}{3} \right] dx$$



$$= \left[\frac{2x^3}{3} - \frac{x^4}{4} + \frac{(2-x)^4}{-12} - \frac{x^4}{3} \right]_0^1$$

$$= \left[\frac{2}{3} - \frac{1}{4} - \frac{1}{12} - \frac{1}{3} \right] - [0]$$

$$= 0$$

Example 4: Evaluate $\iiint xyz\, dxdydz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = 4$.

Solution: let $x = r \sin\theta \cos\varphi$, $y = r \sin\theta \sin\varphi$, $z = r \cos\varphi$

$$I = \iiint xyz\, dxdydz$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^{2} (r \sin\theta \cos\varphi)\,(r \sin\theta \sin\varphi)(r \cos\varphi)\; r^2 \sin\theta \; dr\; d\theta\; d\varphi$$

$$= \int_0^{\frac{\pi}{2}} sin^3\theta \;\cos\theta\; d\theta \;\cdot\; \int_0^{\frac{\pi}{2}} \frac{\sin 2\varphi}{2} \;\cdot\; \int_0^{2} r^5\; dr \qquad \sin 2\varphi = 2\; sin\varphi\; cos\varphi$$

$$= \left[\frac{sin^4\theta}{4}\right]_0^{\frac{\pi}{2}} \cdot \left[\frac{-\cos 2\varphi}{4}\right]_0^{\frac{\pi}{2}} \cdot \left[\frac{r^6}{6}\right]_0^{2} \qquad \left(\because \int [f(\theta)]^n f'(\theta)\, d\theta = \frac{[f(\theta)]^{n+1}}{n+1}\right)$$

$$= \frac{1}{4}\left(sin^4\frac{\pi}{2} - \sin 0\right)\left[\frac{-1}{4}(\cos\pi - \cos 0)\right]\left(\frac{32}{3}\right)$$

$$= \frac{1}{4}(1)\left[\frac{-1}{4}(-2)\right]\left(\frac{32}{3}\right)$$

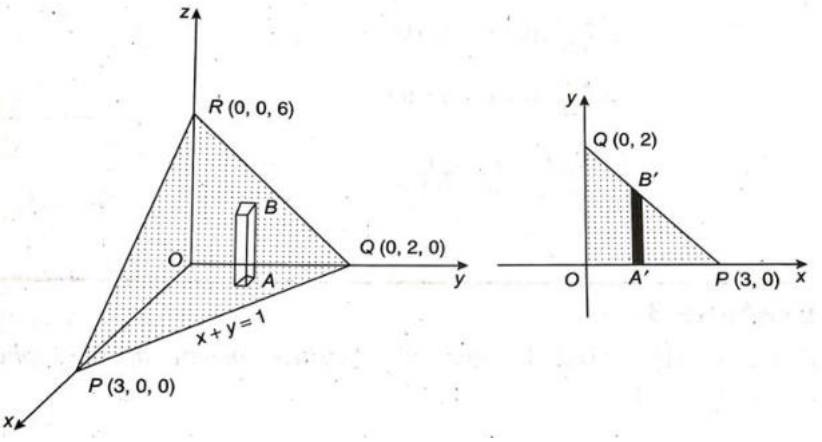
$$= \frac{4}{3}$$

Example 5: Evaluate $\iiint_E \; 2x\; dV$ where E is the region under the plane $2x + 3y + z = 6$ that lies in the first octant.

Solution: limits are $z = 0 \quad to \quad z = 6 - 2x - 3y$

$$y = 0 \quad to \quad y = \frac{6 - 2x}{3}$$

$$x = 0 \quad to \quad x = 3$$



$$I = \iiint_E 2x\, dV$$

$$= \int_0^3 \int_0^{\frac{6-2x}{3}} \int_0^{6-2x-3y} 2x\, dz\, dy\, dx$$

$$= \int_0^3 \int_0^{\frac{6-2x}{3}} 2x\, |z|_0^{6-2x-3y}\, dy\, dx$$

$$= 2\int_0^3 \int_0^{\frac{6-2x}{3}} x(6-2x-3y)\ dy\, dx$$

$$= 2\int_0^3 x \left|(6-2x)y - \frac{3y^2}{2}\right|_0^{\frac{6-2x}{3}} dx$$

$$= 2\int_0^3 x \left[(6-2x)\frac{(6-2x)}{3} - \frac{3}{2}\left(\frac{6-2x}{3}\right)^2\right] dx$$

$$= 2\int_0^3 x\ \frac{(6-2x)^2}{6}\, dx$$

$$= \frac{4}{3}\int_0^3 x\,(3-x)^2\, dx$$

$$= \frac{4}{3}\int_0^3 x\,(9 + x^2 - 6x)\,dx$$

$$= \frac{4}{3}\int_0^3 (9x + x^3 - 6x^2)\,dx$$

$$= \frac{4}{3}\left[9\frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{1}\right]_0^3$$

$$= \frac{4}{3}\left[\frac{81}{2} + \frac{81}{4} - \frac{54}{1}\right] = \frac{4}{3}\left[\frac{27}{4}\right] = 9$$

Example 6: Evaluate $\iiint z^2 dxdydz$ over the region common to the sphere $x^2 + y^2 + z^2 = 4$ and the cylinder $x^2 + y^2 = 2x$.

Solution: The equations in cylindrical coordinates (r, θ, z) are:

$$x = r\cos\theta\,, \quad y = r\sin\theta, \quad z = z$$

$$x^2 + y^2 = r^2$$

The sphere's equation $x^2 + y^2 + z^2 = 4$ becomes

$$r^2 + z^2 = 4$$

$$\therefore z^2 = 4 - r^2$$

The cylinder's equation $x^2 + y^2 = 2x$ becomes:

$$r^2 = 2r\,cos\theta \rightarrow r = 2\,cos\theta$$

Determine the bounds:

- **For r**: The cylinder is bounded by $r = 2\,cos\theta$, so by $0 \le r \le 2\,cos\theta$.
- **For θ**: The cylinder $r = 2\,cos\theta$ exists when $\frac{-\pi}{2} \le \theta \le \frac{\pi}{2}$
- **For z**: The sphere gives $z = \pm\sqrt{4 - r^2}$ so $-\sqrt{4 - r^2} \le z \le \sqrt{4 - r^2}$

Now $I = \iiint z^2 dx\, dy\, dz = \iiint z^2 \;\; r dr\, d\theta\, dz$

$$= \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} z^2 \;\; r\, dz\, dr\, d\theta$$

$$= \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} z^2 \;\; r\, dz\, dr\, d\theta$$

$$= \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} r \;\left|\frac{z^3}{3}\right|_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} dr\, d\theta$$

$$= \frac{1}{3} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} r \left((4-r^2)^{\frac{3}{2}} + (4-r^2)^{\frac{3}{2}}\right) dr\, d\theta$$

$$= \frac{1}{3} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} \;\; 2\, r\, (4-r^2)^{\frac{3}{2}} \;\; dr\, d\theta$$

$$= -\frac{1}{3} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\,cos\theta} (-2\, r)\, (4-r^2)^{\frac{3}{2}}\, dr\, d\theta$$

$$= -\frac{1}{3} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \left|\frac{(4-r^2)^{\frac{5}{2}}}{5/2}\right|_0^{2\,cos\theta} d\theta$$

$$= -\frac{2}{15} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \left[(4-4cos^2\theta)^{\frac{5}{2}} - (4)^{\frac{5}{2}}\right] d\theta$$

$$= -\frac{2}{15} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \left[(4)^{\frac{5}{2}}(sin^2\theta)^{\frac{5}{2}} - (4)^{\frac{5}{2}}\right] d\theta$$

$$= -\frac{2}{15} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} 2^5\, [(sin^5\theta) - 1]\, d\theta$$

$$= -\frac{2^6}{15} \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} [(sin^5\theta) - 1]\, d\theta$$

$$= -\frac{2^6}{15}\left[0 - |\theta|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\right] \qquad since \int_{-a}^{a} f(x)\,dx = 0,$$

$$if\ f(x)\ is\ ODD\ function\ here\ sin\theta\ is\ odd$$

$$= -\frac{2^6}{15}\left[-\left(\frac{\pi}{2} + \frac{\pi}{2}\right)\right]$$

$$= \frac{64\,\pi}{15}$$

Example 7: Use triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ between the planes $z = 1 and\ x + z = 1$.

Solution: For Cylinder $x^2 + y^2 = 9$ and the planes $x + z = 1$.

$$x = \pm\sqrt{9 - y^2} \text{ and } z = 1 - x$$

$\therefore$ z varies from $z = 0$ to $z = 1 - x$

x varies from $x = -\sqrt{9 - y^2}$ to $x = \sqrt{9 - y^2}$

y varies from $y = -3$ to $y = 3$

$\therefore$ Required volume $V = \iiint dx\,dy\,dz$

$$= \int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}\int_{0}^{1-x} dz\,dx\,dy$$

$$= \int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} (1 - x)\quad dx\,dy$$

$$= \int_{-3}^{3}\left[x - \frac{x^2}{2}\right]_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \quad dx\,dy$$

$$= \int_{-3}^{3} \left[\sqrt{9-y^2} - \frac{1}{2}(9-y^2) + \sqrt{9-y^2} + \frac{1}{2}(9-y^2) \right] dy$$

$$= 2\int_{-3}^{3} \sqrt{9-y^2}\ dy = 4\int_{0}^{3} \sqrt{9-y^2}\ dy \quad \left(\because \sqrt{9-y^2}\ is\ odd\ function\right)$$

$$= 4\left[\frac{y}{2}\sqrt{9-y^2} + \frac{9}{2}\sin^{-1}\frac{y}{3}\right]_0^3$$

$$= 4\left[0 + \frac{9}{2}\sin^{-1}1 - 0 - 0\right]$$

$$= 4\left[\frac{9}{2}\frac{\pi}{2}\right] = 9\pi$$

Example 8: Use triple integral to find the volume of the cylinder $x^2 + y^2 = 1$ between the places z=1 and z=2.

Solution:

- The equation of the cylinder $x^2 + y^2 = 1$ translates to $r = 1$ in cylindrical coordinates, where r is the radial distance.
- The height of the cylinder is between $z = 1$ and $z = 2$
- The angular coordinate θ will range from 0 to 2π

So, the bounds for the cylindrical coordinates (r, θ, z) are:

$$0 \le r \le 1$$

$$0 \le \theta \le 2\pi$$

$$1 \le z \le 2$$

$$\therefore \text{The Volume } V = \int_0^{2\pi} \int_0^1 \int_1^2 r\ dz\ dr\ d\theta$$

$$= \int_0^{2\pi} \int_0^1 r\ |z|_1^2\ dr\ d\theta$$

$$= \int_0^{2\pi} \int_0^1 r\ \ dr\ d\theta$$

$$= \int_0^{2\pi} \left| \frac{r^2}{2} \right|_0^1 d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} (1 - 0) \; d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} 1 \; d\theta$$

$$= \frac{1}{2} |\theta|_0^{2\pi}$$

$$= \frac{1}{2}(2\pi) = \pi$$

Example 9: Find the volume of the tetrahedron bounded by the plane $x + y + z = 2$ and the planes $x = 0, y = 0 \; and \; z = 0$.

Solution:

- For a fixed x and y, z ranges from 0 to $2 - x - y$
- For a fixed x, y ranges from 0 to $2 - x$
- Finally, x ranges from 0 to 2.

The volume V of the tetrahedron can be calculated as:

$$V = \int_0^2 \int_0^{2-x} \int_0^{2-x-y} dz \, dy \, dx$$

$$= \int_0^2 \int_0^{2-x} |z|_0^{2-x-y} \; dy \, dx$$

$$= \int_0^2 \int_0^{2-x} |z|_0^{2-x-y} \; dy \, dx$$

$$= \int_0^2 \int_0^{2-x} (2 - x - y) \; dy \, dx$$

$$= \int_0^2 \left| 2y - xy - \frac{y^2}{2} \right|_0^{2-x} \quad dx$$

$$= \int_0^2 \left[2(2-x) - x(2-x) - \frac{(2-x)^2}{2} \right] \quad dx$$

$$= \int_0^2 \left[4 - 2x - 2x + x^2 - \frac{(4 - 4x + x^2)}{2} \right] \quad dx$$

$$= \int_0^2 \left[\frac{x^2 - 4x + 4}{2} \right] \quad dx$$

$$= \frac{1}{2}\left[\frac{x^3}{3} - 2x^2 + 4x\right]_0^2 = \frac{1}{2}\left[\frac{8}{3} - 8 + 8\right] = \mathbf{\frac{4}{3}}$$

Example 10 : Use triple integral in cylindrical co-ordinate to find the volume of solid, bounded above by the hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by xy-plane and laterally by the cylinder $x^2 + y^2 = 9$

Solution: The equations in cylindrical coordinates (r, θ, z) are:

$$x = r\cos\theta\,, \quad y = r\sin\theta, \quad z = z$$

$$x^2 + y^2 = r^2$$

$\therefore$ hemisphere $z = \sqrt{25 - x^2 - y^2}$ becomes $z = \sqrt{25 - r^2}$ and

the cylinder $x^2 + y^2 = 9$ becomes $r^2 = 3$

So, the bounds for the cylindrical coordinates (r, θ, z) are:

$0 \le r \le 3$

$0 \le \theta \le 2\pi$

$0 \le z \le \sqrt{25 - r^2}$

$\therefore$ The Volume $V = \int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r\, dz\, dr\, d\theta$

$$= \int_0^{2\pi} \int_0^3 r\ |z|_0^{\sqrt{25-r^2}}\ dr\ d\theta$$

$$= \int_0^{2\pi} \int_0^3 r\ \left(\sqrt{25-r^2}\right) dr\ d\theta$$

$$= \int_0^{2\pi} \int_0^3 r\ \left((25-r^2)^{\frac{1}{2}}\right) dr\ d\theta$$

$$= -\frac{1}{2}\int_0^{2\pi} \int_0^3 (-2r)\ \left((25-r^2)^{\frac{1}{2}}\right) dr\ d\theta$$

$$= -\frac{1}{2}\int_0^{2\pi} \left|\frac{(25-r^2)^{\frac{3}{2}}}{3/2}\right|_0^3\ d\theta$$

$$= -\frac{1}{3}\int_0^{2\pi} \left[(16)^{\frac{3}{2}} - (25)^{\frac{3}{2}}\right]\ d\theta$$

$$= -\frac{1}{3} \times (64-125) \int_0^{2\pi} 1\ d\theta$$

$$= -\frac{1}{3} \times (-61)\ [\theta]_0^{2\pi}$$

$$= \frac{1}{3} \times (61) \times 2\pi\ .$$

$$= \frac{122\ \pi}{3}$$

Example 11 : Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$

Solution: The equation of the sphere $x^2 + y^2 + z^2 = a^2$ can be converted to spherical coordinates (r, θ, φ) where

$$x = r\ \sin\theta\ \cos\varphi$$

$$y = r\ \sin\theta\ \sin\varphi$$

$z = r\ \cos\theta$

The equation $x^2 + y^2 + z^2 = a^2$ becomes $r^2 = a^2$ and so $r = a$

So, the bounds are:

$0 \le r \le a \qquad 0 \le \theta \le \pi \qquad 0 \le \varphi \le 2\pi$

$$\therefore \text{The Volume } V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^{\,2}\sin\theta\ dr\ d\theta\ d\varphi$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\theta\ \left|\frac{r^3}{3}\right|_0^a d\theta\ d\varphi$$

$$= \frac{a^3}{3}\int_0^{2\pi}\int_0^{\pi} \sin\theta\ d\theta\ d\varphi$$

$$= \frac{a^3}{3}\int_0^{2\pi} |-\cos\theta|_0^{\pi}\ d\varphi$$

$$= \frac{a^3}{3}\int_0^{2\pi} -(\cos\pi - \cos 0)\ d\varphi$$

$$= \frac{2a^3}{3}\int_0^{2\pi} 1\ d\varphi$$

$$= \frac{2a^3}{3}\ |\varphi|_0^{2\pi}$$

$$= \frac{2a^3}{3}(2\pi)$$

$$= \frac{4\pi a^3}{3}$$

Example 12: Find the volume bounded by the cylinder $x^2 + z^2 = 1, y = 0$ and $y + z = 3$ using triple integration.

Solution: let the cylindrical coordinates $x = r\ \cos\theta$

$$z = r\ \sin\theta$$

$$y = y$$

The cylinder $x^2 + z^2 = 1$ becomes $r^2 = 1$ so r ranges from 0 to 1

Angle θ ranges from $0 \; to \; 2\pi$

y ranges from 0 to $3 - z = 3 - r \sin\theta$

$$\therefore \text{The Volume } V = \int_0^{2\pi} \int_0^1 \int_0^{3-r\sin\theta} r \; dy \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 r \, |y|_0^{3-r\sin\theta} \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 r \, (3 - r \sin\theta) \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 (3r - r^2 \sin\theta) \, dr \, d\theta$$

$$= \int_0^{2\pi} \left| \frac{3r^2}{2} - \sin\theta \frac{r^3}{3} \right|_0^1 d\theta$$

$$= \int_0^{2\pi} \left(\frac{3}{2} - \frac{\sin\theta}{3} \right) d\theta$$

$$= \left| \frac{3}{2}\theta + \frac{\cos\theta}{3} \right|_0^{2\pi}$$

$$= \left(3\pi + \frac{1}{3} - 0 - \frac{1}{3} \right)$$

$$= 3\pi$$

Example 13 : Evaluate $\iiint z(x^2 + y^2) dv$ over the volume of the cylinder $x^2 + y^2 = 1$ intercepted by the planes z =2 and z = 3.

Solution: In cylindrical coordinates we have $x = r\ \cos\theta$

$$y = r\ \sin\theta$$

$$z = z$$

Where r is the radial distance, θ is the angular coordinate, and z is the height.

The equation of the cylinder $x^2 + y^2 = 1$ becomes $r^2 = 1$ and so $r = 1$

$$\therefore I = \iiint z(x^2 + y^2)dv = \int_2^3 \int_0^1 \int_0^{2\pi} z\, r^2 \quad r\, d\theta\, dr\, dz$$

$$= \int_2^3 \int_0^1 z\, r^3 \ \ |\theta|_0^{2\pi}\ dr\, dz$$

$$= 2\pi \int_2^3 \int_0^1 z\, r^3 \quad dr\, dz$$

$$= 2\pi \int_2^3 z \ \left|\frac{r^4}{4}\right|_0^1 \ dz$$

$$= 2\pi \int_2^3 z\ \frac{1}{4}\ dz$$

$$= \frac{\pi}{2} \int_2^3 z \quad dz$$

$$= \frac{\pi}{2} \left|\frac{z^2}{2}\right|_2^3$$

$$= \frac{\pi}{4}(9 - 4)$$

$$= \frac{5\,\pi}{4}$$